

Rigid Motions on the Coordinate Plane

Answer Key

Geometry

Quick Answers

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|-----------|---------------------|---------|
| 1. 2, -2 | 4. -1, -4, 3, -1, 5 | 7. 6, 3 |
| 2. -6, -2 | 5. 4, 1, 7, 5, 5 | 8. — |
| 3. -7, 2 | 6. 3, -5, -1, 4 | |
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Curriculum

Level: Geometry

HSG-CO.A–HSG-CO.D – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

7. *If they got -6: reflected over the x-axis, which leaves the x-coordinate alone instead of negating it*
7. *If they got -3: reflected over the x-axis, which negates the y-coordinate*
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What this sheet is teaching. A rigid motion is defined by what it *preserves*, not by what it looks like: distance survives, and therefore so do angle measure and congruence. The sheet keeps the algebra and the picture side by side on every problem so the student can catch a slip in either one. The distance-formula checks on problems 4, 5 and 6 are not busywork — they are the evidence for the congruence claim the student is asked to make in problem 8 without measuring anything.

1. Translate a point. $P(-4, 1)$ under $(x, y) \rightarrow (x + 6, y - 3)$.

Apply the rule coordinate by coordinate:

$$x' = -4 + 6 = \boxed{2}, \quad y' = 1 - 3 = \boxed{-2}.$$

So $P'(2, -2)$: six units right and three units down from P .

Check on the grid. Counting from P , six squares right and three squares down lands in the fourth quadrant, which is where the arithmetic put it. A student whose plotted point disagrees with their algebra has almost always counted the negative direction the wrong way.

2. Reflect a point over the y -axis. $Q(6, -2)$.

The y -axis is a vertical mirror, so it can only move a point left or right: the height above or below the x -axis is untouched. The rule is $(x, y) \rightarrow (-x, y)$.

$x' = -6$, so $x' = \boxed{-6}$; $y' = \boxed{-2}$, unchanged.

So $Q'(-6, -2)$.

The distance check. Q is 6 units right of the y -axis and Q' is 6 units left of it. That equal-distance-on-opposite-sides property *is* the definition of a reflection; the coordinate rule is just its shorthand.

Common slip: negating both coordinates. That is a 180° rotation, not a reflection, and it is the same confusion problem 7 is built around.

3. Rotate $R(2, 7)$ by 90° counterclockwise about the origin.

The rule is $(x, y) \rightarrow (-y, x)$. Swap the coordinates, then negate the new first one:

$x' = -7$, so $x' = \boxed{-7}$; $y' = \boxed{2}$.

So $R'(-7, 2)$.

Two checks the student can run alone. The spoke OR and the spoke OR' must be the same length: $\sqrt{2^2 + 7^2} = \sqrt{53}$ both times, because swapping and negating does not change the sum of the squares. And they must be perpendicular — the slopes are $7/2$ and $-2/7$, whose product is -1 .

Direction matters. A clockwise quarter turn is $(x, y) \rightarrow (y, -x)$ and would give $(7, -2)$. If a student produces that, they have the mechanism right and the direction wrong; ask which way the dashed spoke has to sweep.

4. Translate the segment $A(-5, 2)$, $B(-1, 5)$ by $(x, y) \rightarrow (x + 4, y - 6)$.

The image points.

$A' = (-5 + 4, 2 - 6) = (\boxed{-1}, \boxed{-4})$, $B' = (-1 + 4, 5 - 6) = (\boxed{3}, \boxed{-1})$.

The length check. Using the distance formula on the image:

$$A'B' = \sqrt{(3 - (-1))^2 + (-1 - (-4))^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = \boxed{5}.$$

And on the pre-image: $AB = \sqrt{(-1 - (-5))^2 + (5 - 2)^2} = \sqrt{16 + 9} = 5$. The two agree, as they must.

Why it had to work. Both endpoints moved by the same vector, so the difference between them — which is all the distance formula sees — did not change at all. This is the cleanest of the three rigid motions to argue: the horizontal gap of 4 and the vertical gap of 3 are literally the same numbers before and after.

5. Reflect the segment $C(1, 4)$, $D(5, 7)$ over the line $y = x$.

The image points. The rule $(x, y) \rightarrow (y, x)$ swaps the coordinates:

$$C' = (\boxed{4}, \boxed{1}), \quad D' = (\boxed{7}, \boxed{5}).$$

The length check.

$$C'D' = \sqrt{(7-4)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25} = \boxed{5},$$

and $CD = \sqrt{(5-1)^2 + (7-4)^2} = \sqrt{16+9} = 5$. Equal.

Look at what the swap did to the formula. The horizontal gap and the vertical gap traded places — 4 and 3 became 3 and 4 — and the distance formula squares both and adds, so the order cannot matter. That is a proof, not a coincidence, and it works for any segment reflected in $y = x$.

The picture check. The mirror line should be the perpendicular bisector of $\overline{CC'}$. Its midpoint is $(2.5, 2.5)$, which lies on $y = x$, and the segment from $(1, 4)$ to $(4, 1)$ has slope -1 , perpendicular to the mirror's slope of 1.

6. Rotate the segment $E(-3, 5)$, $F(-3, 1)$ by 180° about the origin.

The image points. A half turn negates both coordinates, $(x, y) \rightarrow (-x, -y)$:

$$E' = (\boxed{3}, \boxed{-5}), \quad F' = (3, \boxed{-1}).$$

The length check. Both image points share the x -coordinate 3, so the segment is vertical and the length is just the gap in y :

$$E'F' = \sqrt{(3-3)^2 + (-1-(-5))^2} = \sqrt{0+16} = \boxed{4}.$$

The pre-image $\overline{EF}$ is also vertical, from $y = 1$ to $y = 5$, so $EF = 4$.

The quadrant answer. $\overline{EF}$ sits in the second quadrant and its image sits in the fourth — diagonally opposite, which is what a half turn about the origin always does. Note that the segment stayed vertical: a 180° rotation preserves every direction up to reversal, which is why it is the one rotation students can do in their heads.

7. Find and fix the mistake. Reflect $G(-6, 3)$ over the y -axis; Sam wrote $G'(-6, -3)$.

What Sam did. Sam left the x -coordinate alone and negated the y -coordinate. That is the rule $(x, y) \rightarrow (x, -y)$ — a reflection over the x -axis, not the y -axis.

The correct image. Reflecting over the y -axis uses $(x, y) \rightarrow (-x, y)$:

$$x' = -(-6) = \boxed{6}, \quad y' = \boxed{3} \text{ (unchanged).}$$

So $G'(6, 3)$.

How to remember which is which. Name the mirror and ask which way it points. The y -axis is vertical, so it flips a point *sideways*, and sideways is the x -coordinate. The x -axis is horizontal, so it flips a point *up and down*, which is the y -coordinate. The axis you reflect over is the one coordinate that does *not* change — students who memorise a pair of rules with no picture attached reliably swap them under time pressure.

On the grid. G sits in the second quadrant. Sam's answer stays on the left, in the third quadrant; the correct image crosses to the right, into the first. A quick plot would have caught it.

8. Name the motion, then justify the congruence. (Open description — read the reasoning, not just the name.)

(a) The motion. It is a 90° **counterclockwise rotation about the origin**, coordinate rule $(x, y) \rightarrow (-y, x)$. Describing it as a 270° clockwise rotation about the origin is the same motion and equally correct.

(b) Checking every vertex.

- $J(2, 1) \rightarrow (-1, 2) = J'$
- $K(6, 1) \rightarrow (-1, 6) = K'$
- $L(6, 4) \rightarrow (-4, 6) = L'$

All three land on the marked image points, so one rule accounts for the whole triangle — which is what “a single rigid motion” means.

(c) The congruence argument. A rotation is a rigid motion, and a rigid motion preserves distance by definition. So $JK = J'K'$, $KL = K'L'$ and $LJ = L'J'$ without computing any of them, and the triangles are congruent by SSS. The point of the question is that congruence follows from the *kind* of transformation, not from an inventory of measurements.

Marking guide. Full credit needs all three parts: the motion named (rotation, 90° , counterclockwise, about the origin), the rule verified on each vertex, and a justification that appeals to rotations preserving distance. Partial credit for a correct name with the vertices checked but the congruence argued by measuring all three sides — correct mathematics, but it is the argument the question asked them to replace. Not sufficient: “they look the same size and shape”.

If a student says translation: ask them what single slide could take a triangle with a horizontal base to one with a vertical base. Orientation of the sides, not position, is what rules a translation out here.

Open description and justification — open response.